SD5531 Lab Study 1 (System Design)

Dr. Clifford CHOY

mccliff@polyu.edu.hk

4 Sep 2026

Ground Rules

- Be punctual
  - Take attendance within the first 15 minutes
- Don’t disturb others during classes
  - No mobile phone, no pager, no eating, no drinking
  - No snoring
- No **plagiarism**!!! Serious offence!!!
- Actively participate
  - Interrupt me when you don’t understand
  - Learn to question, not just listen
  - Think critically

Objectives

- Provide students with a practical and experimental platform to experience hands-on development process of a functional system which interacts with people and/or environment

- Develop student's ability to see systems as a network of interacting components/modules, and appreciate technical characteristics of different types of components/modules (specifically relating to mechanical, electronics and information and computation aspects), and their implications to overall system performance and impact on actual use to people and/or environment

Teaching and Supporting Staff

- Dr. Clifford Choy (Subject Coordinator)
  - Room: V902g
  - email: mccliff@polyu.edu.hk
  - Office: 27664738
  - Whatsapp: 90296782
  - WeChat: CliffordChoy
- Mr John Mak
  - Room: V040
  - email: ty-john.mak@polyu.edu.hk

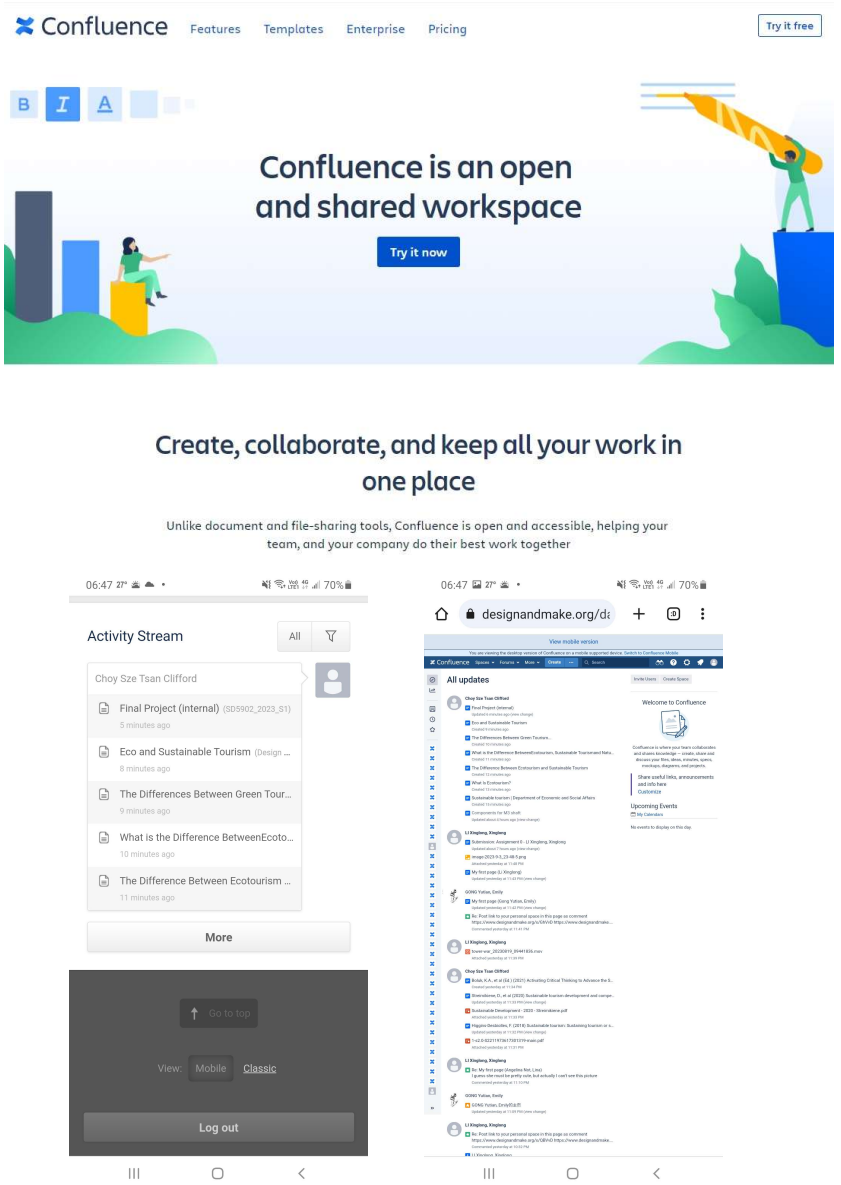
Class Time

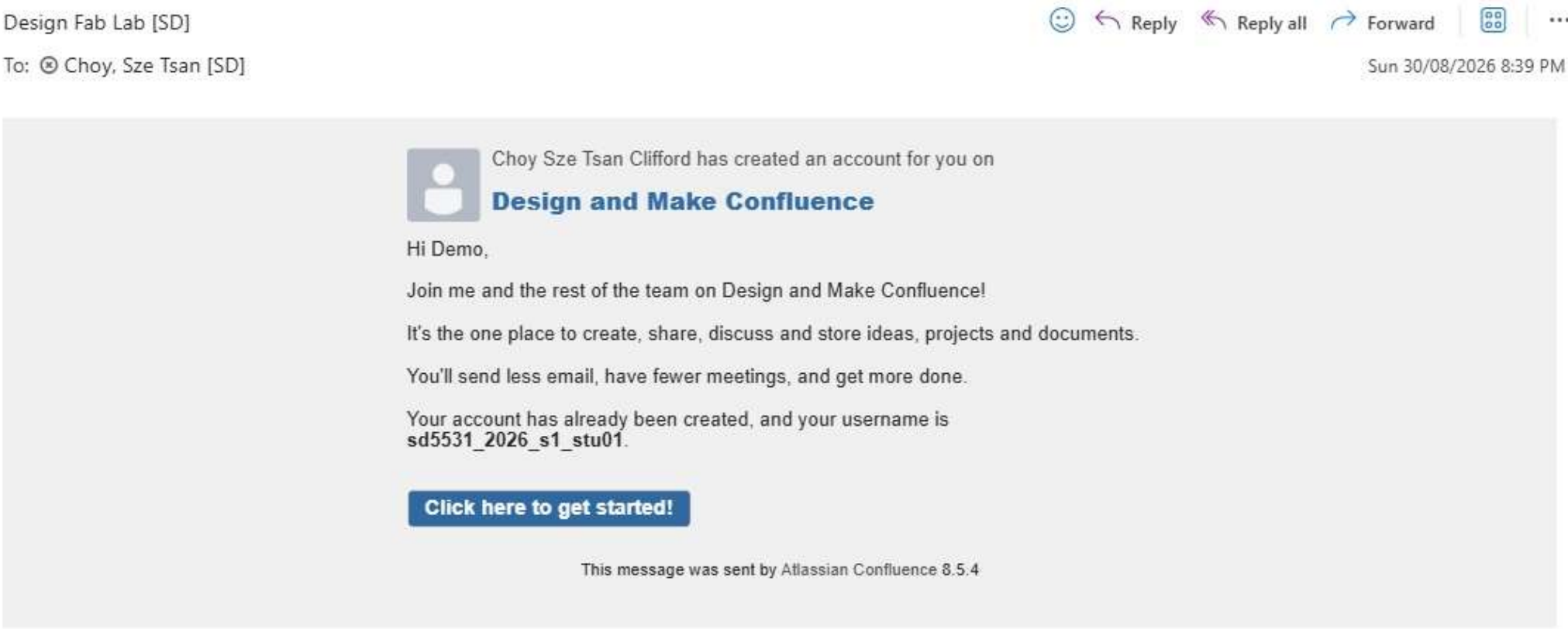
- Lectures/Tutorials
  - Fri 2:30pm – 5:30pm

- Consultation time
  - Wed/Thur 9:30am till 6:30pm
  - Email/whatsapp me at least one day before, as I may have other appointments

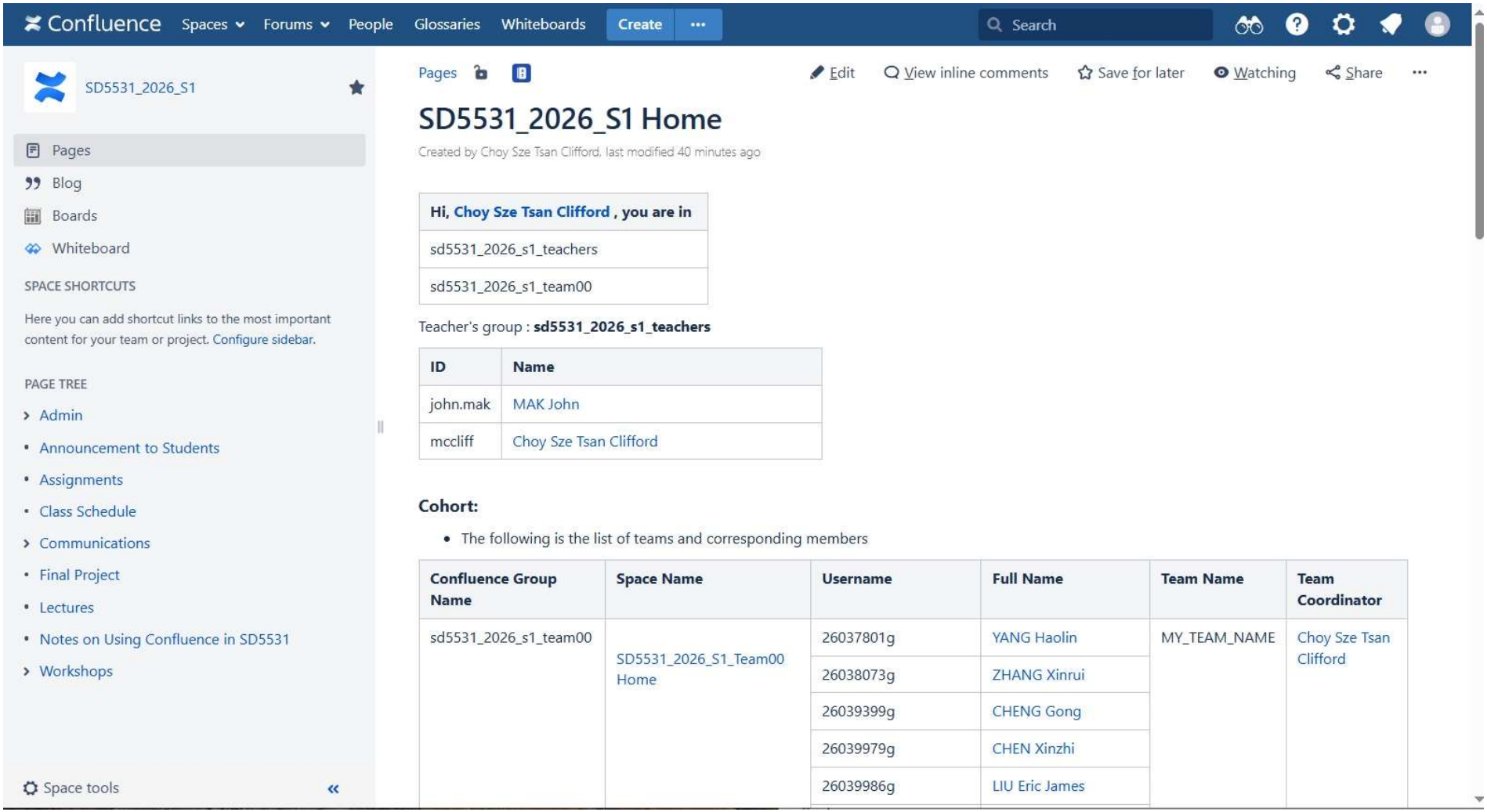
Notes

- As we will be using a Collaborative Platform extensively (Confluence, https://www.designandmake.org), please use a notebook/desktop computer, running Windows or MacOSX is preferable.
- **Not preferable to use mobile phone or tablet**
  - **Use “Desktop Mode” when using mobile devices**
- When in face-to-face meeting, please bring a notebook computer with you to the lectures/tutorials, at least one per team

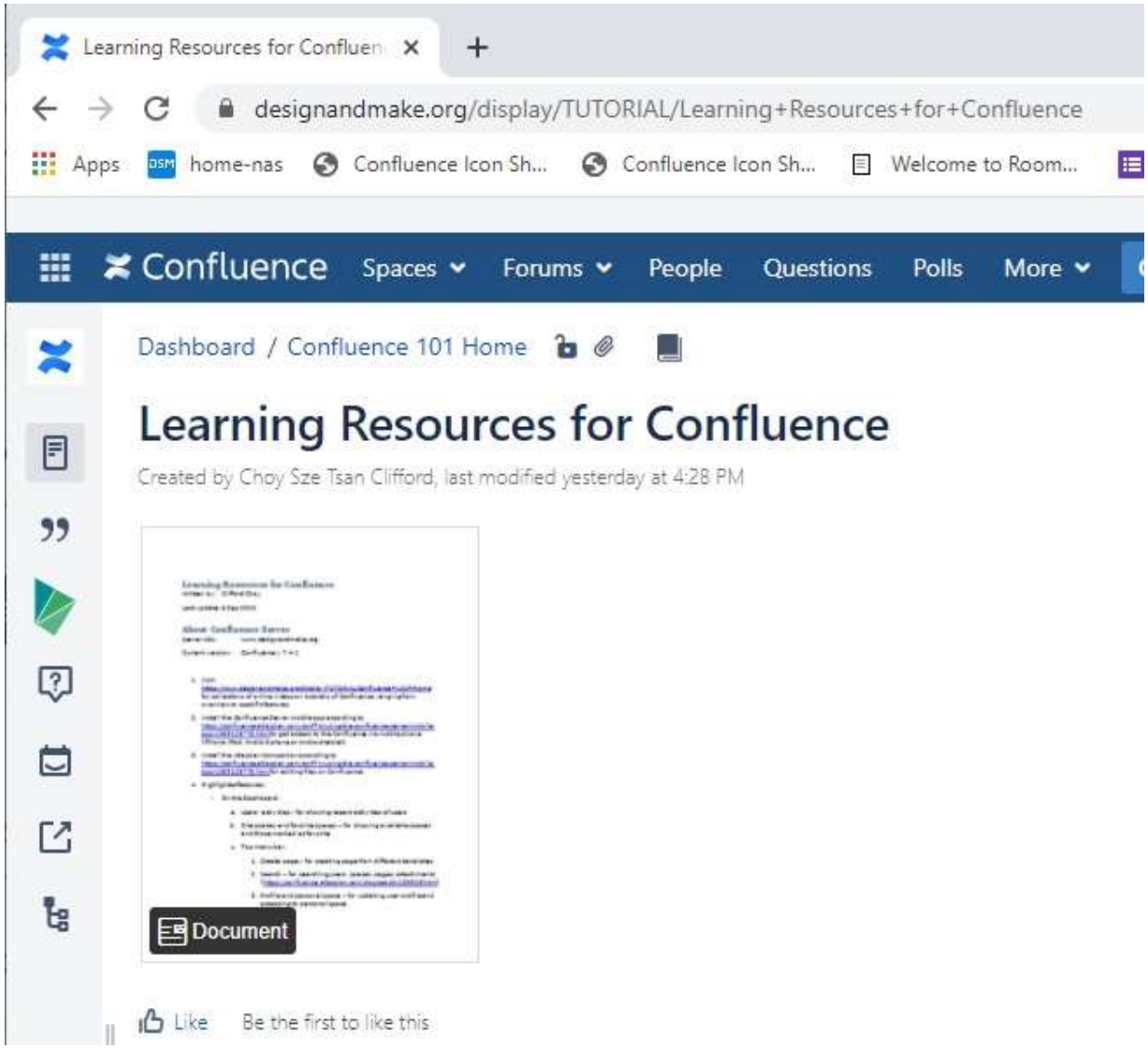




- You should receive an email like this. This may be in the junk email folder.
- Click on the link to setup your account's password
- You user name is your student id



https://www.designandmake.org/x/toT0Eg



https://www.designandmake.org/x/FyN3Aw

Confluence Spaces Forums People Glossaries Whiteboards Create ... Search

SD5531_2026_S1

- Pages
- Blog
- Boards
- Whiteboard

SPACE SHORTCUTS

Here you can add shortcut links to the most important content for your team or project. Configure sidebar.

PAGE TREE

- Admin
- Announcement to Students
- Assignments
- Class Schedule
- Communications
- Final Project
- Lectures
- **Notes on Using Confluence in SD5531**
- Workshops

Space tools

Pages / SD5531_2026_S1 Home Edit View inline comments Save for later Watching Share ...

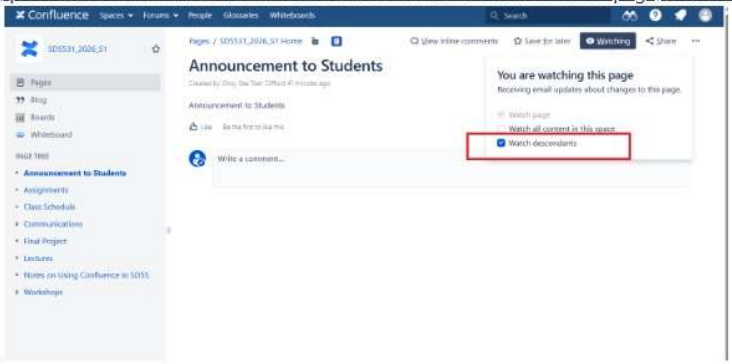
Notes on Using Confluence in SD5531

Created by Choy Sze Tsan Clifford, last modified 2 minutes ago

Different groups:

1. teachers are in the group **sd5531_2026_s1_teachers**
2. all students are in the group **sd5531_2026_s1_students**
3. group names for different teams are in SD5531_2026_S1 Home

How to communicate:

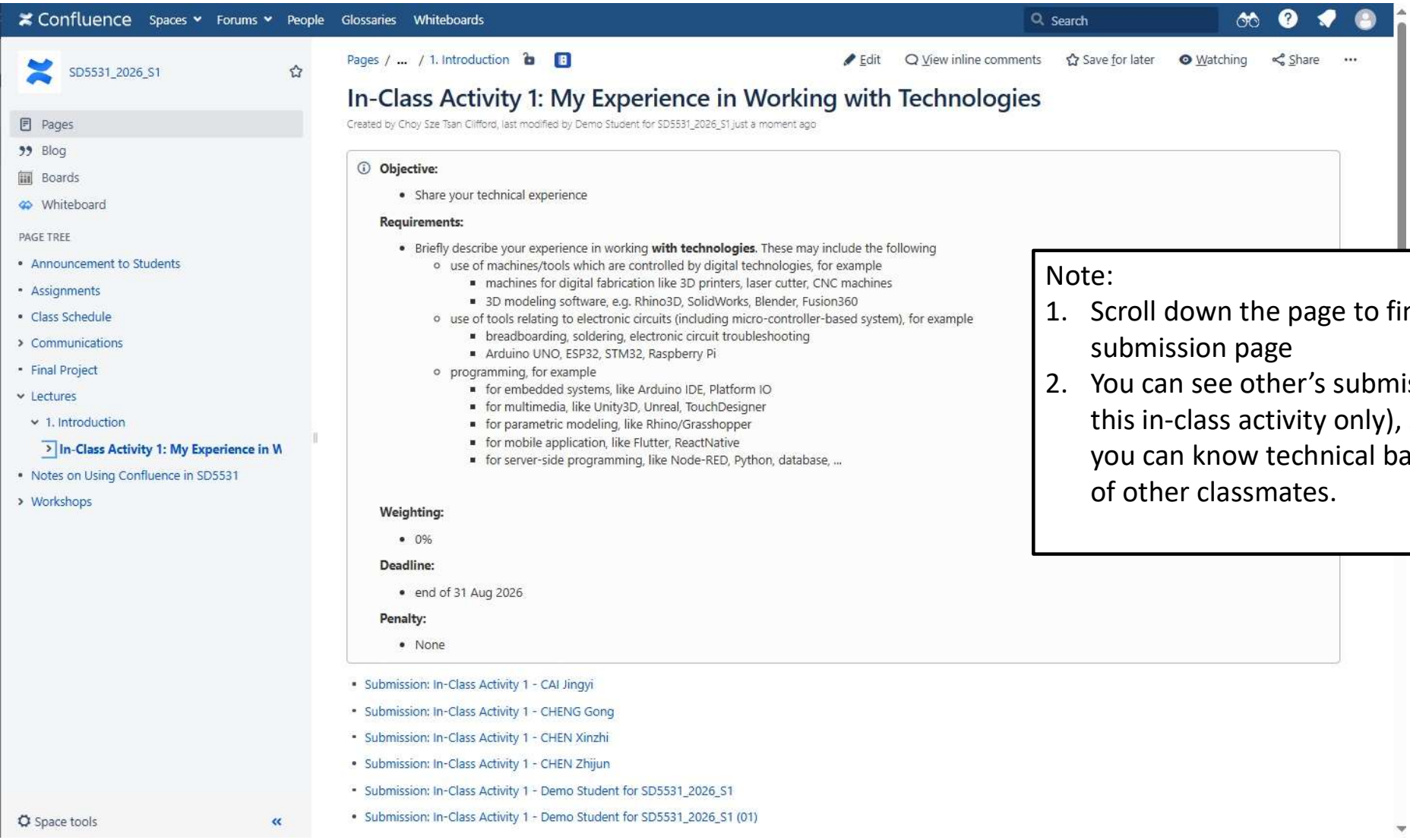
1. teacher(s) → all students
   a. message will be sent in Announcement to Students (either as a comment in the page, or in a page created under this page)
   b. Hence, please be reminded to check **Watch descendants** to announcement page, and check email frequently
      i. 

2. teacher(s) → individual student
   a. message will sent as comments to the page under Communications, or as page under it
   b. Hence, please be reminded to check **Watch descendants** to the page

https://www.designandmake.org/x/A4X0Eg

Trial of Selected Functions

- Watch
  - Announcement page
  - Communication channels
- Create a page in SD5531_2026_S1_Team00 under “Public/Test Page 1” (https://www.designandmake.org/x/Q4X0Eg)
  - Save after adding a title, before making any change
  - No two pages with the same name within the same space
- Add video to page (using “VideoJS” macro)
- Auto-save
- Shared editing
- History
- Mention, notifications
- Comments
  - Below page content
  - Inline
- Move pages, from under “Public/Test Page 1” to under “Public/Test Page 2”
- Copy
- Set restrictions



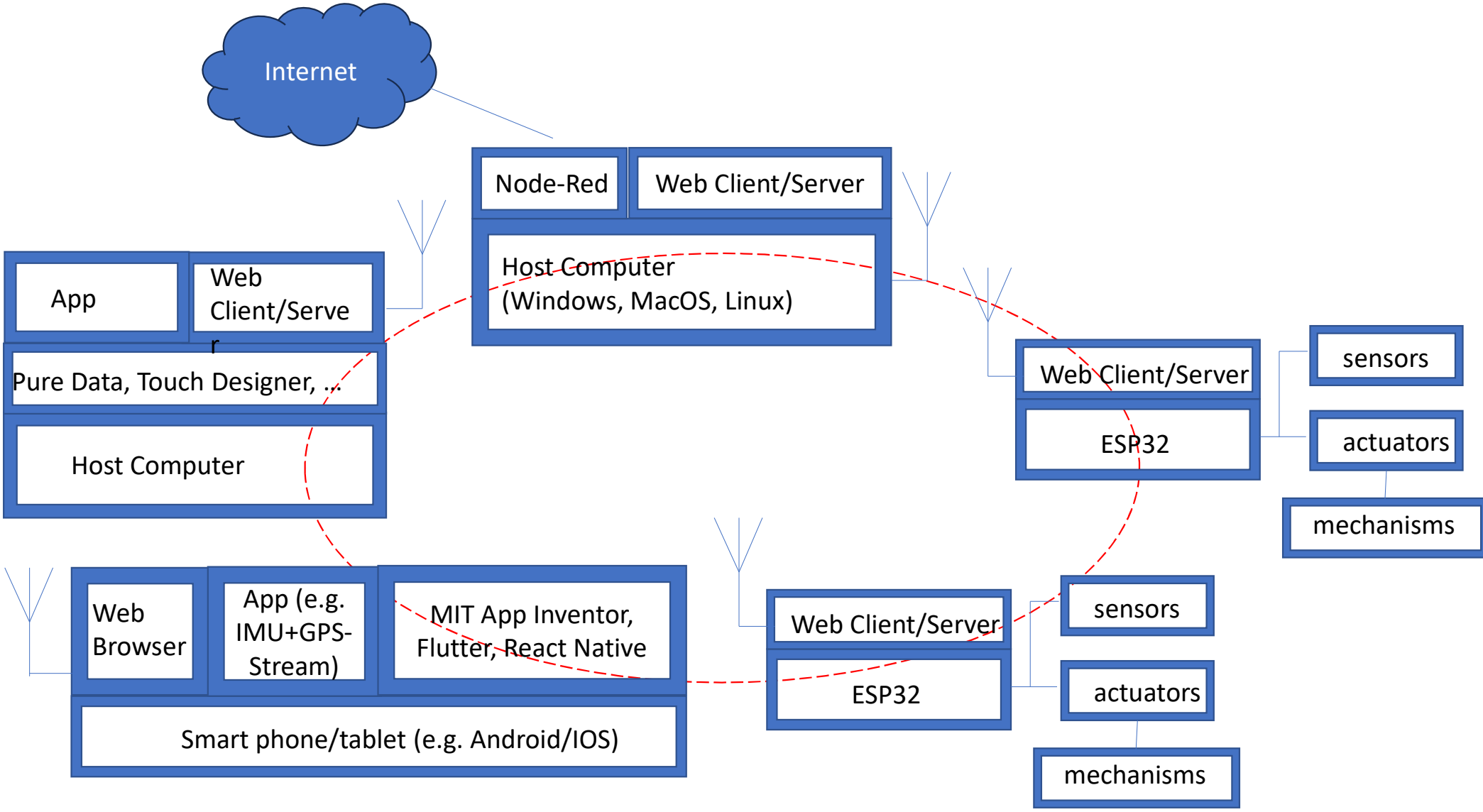
Note:

1. Scroll down the page to find you submission page
2. You can see other's submission (for this in-class activity only), so that you can know technical background of other classmates.

https://www.designandmake.org/x/GYX0Eg

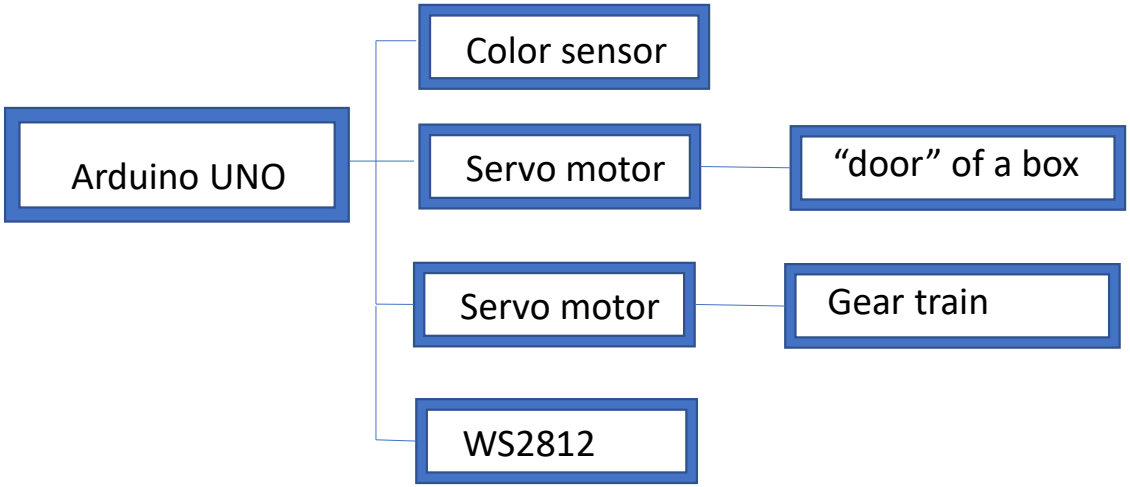
10min

- Try those selected functions on use of Confluence
- Work on In-class Activity 1

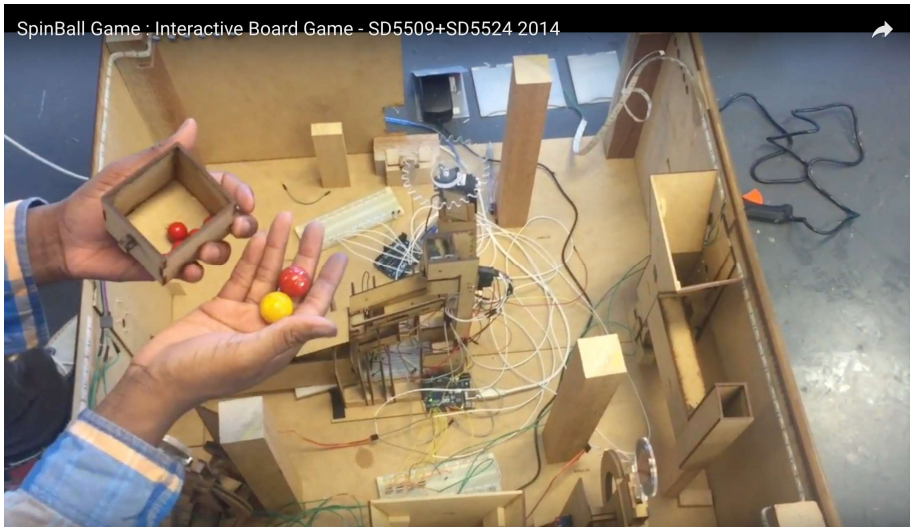

Internet
Node-Red
Web Client/Server
Host Computer
(Windows, MacOS, Linux)
App
Web
Client/Serve
r
Pure Data, Touch Designer, ...
Host Computer
Web Client/Server
ESP32
sensors
actuators
mechanisms
Web
Browser
App (e.g.
IMU+GPS-
Stream)
MIT App Inventor,
Flutter, React Native
Smart phone/tablet (e.g. Android/IOS)
Web Client/Server
ESP32
sensors
actuators
mechanisms

Previous Project Works

- Arduino UNO
- Color sensor
- Servo motor
- DC motor
- LEDs (NeoPixel/WS2812)

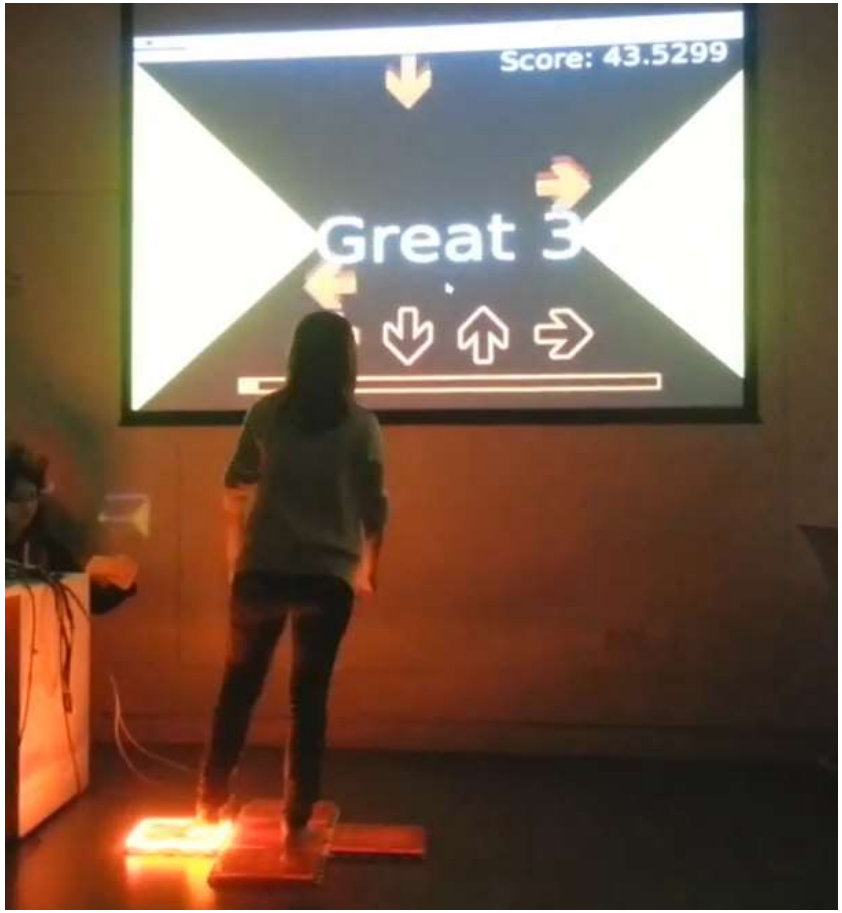






https://youtu.be/4UnCLJMSChQ

- Arduino UNO
  - Piezoelectric sensor
  - LEDs
- Pure Data running on PC
  - Communicate with Arduino UNO via USB
  - Video playback
  - Update score on screen

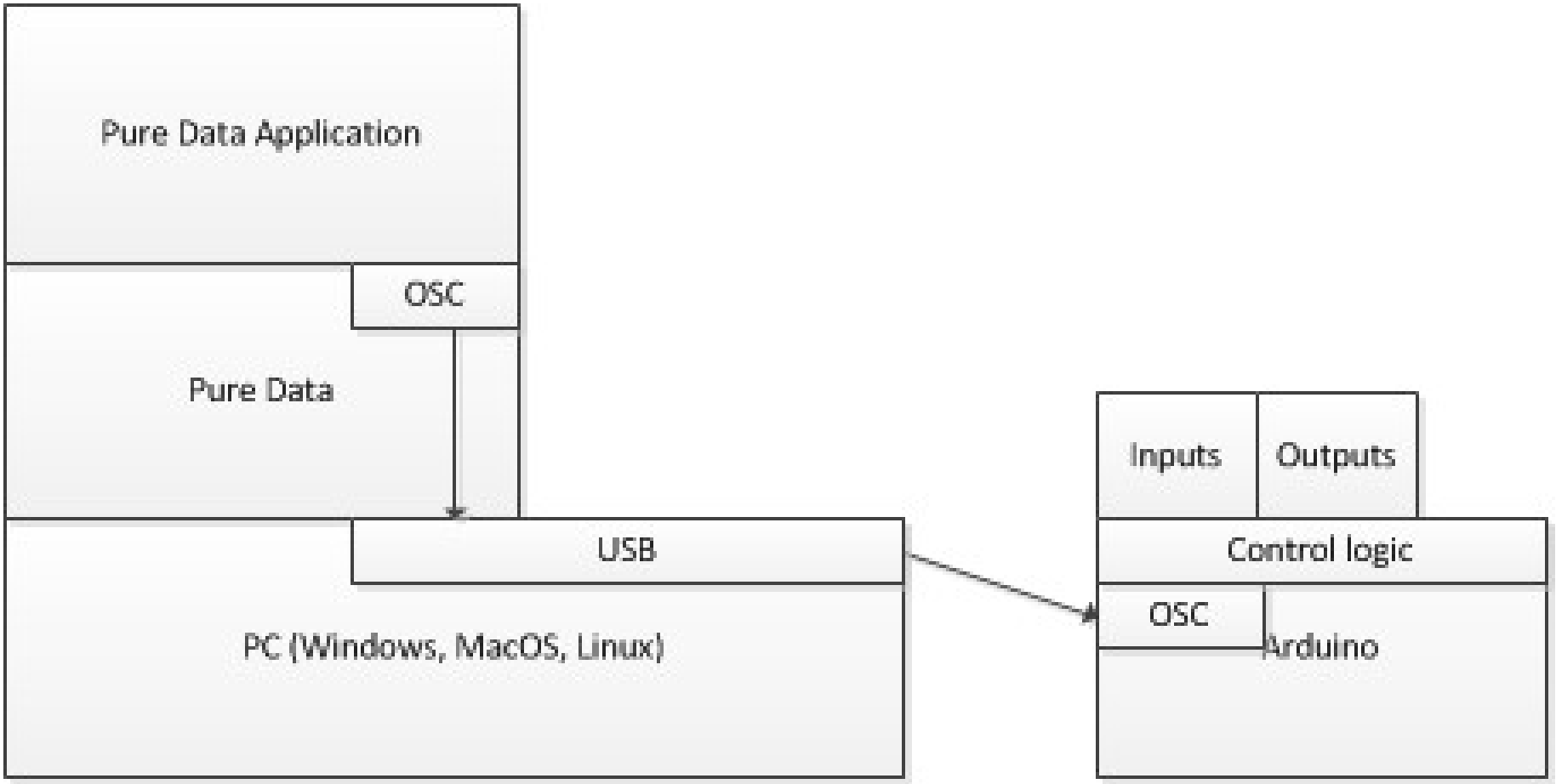


https://youtu.be/FXBe26N7pEw
(https://www.designandmake.org/x/tgLJ)

- Arduino UNO
  - Color sensor
  - Accelerometer
  - LEDs
- Bluetooth speaker
- Pure Data running on PC
  - Music generation
  - Communicate with Arduino UNO



https://www.designandmake.org/display/shareit/13795982/GZA0148e2c231c54884b7917078b8038704QOV/duckdemovideo.mp4?version=1 (https://www.designandmake.org/x/joLS)

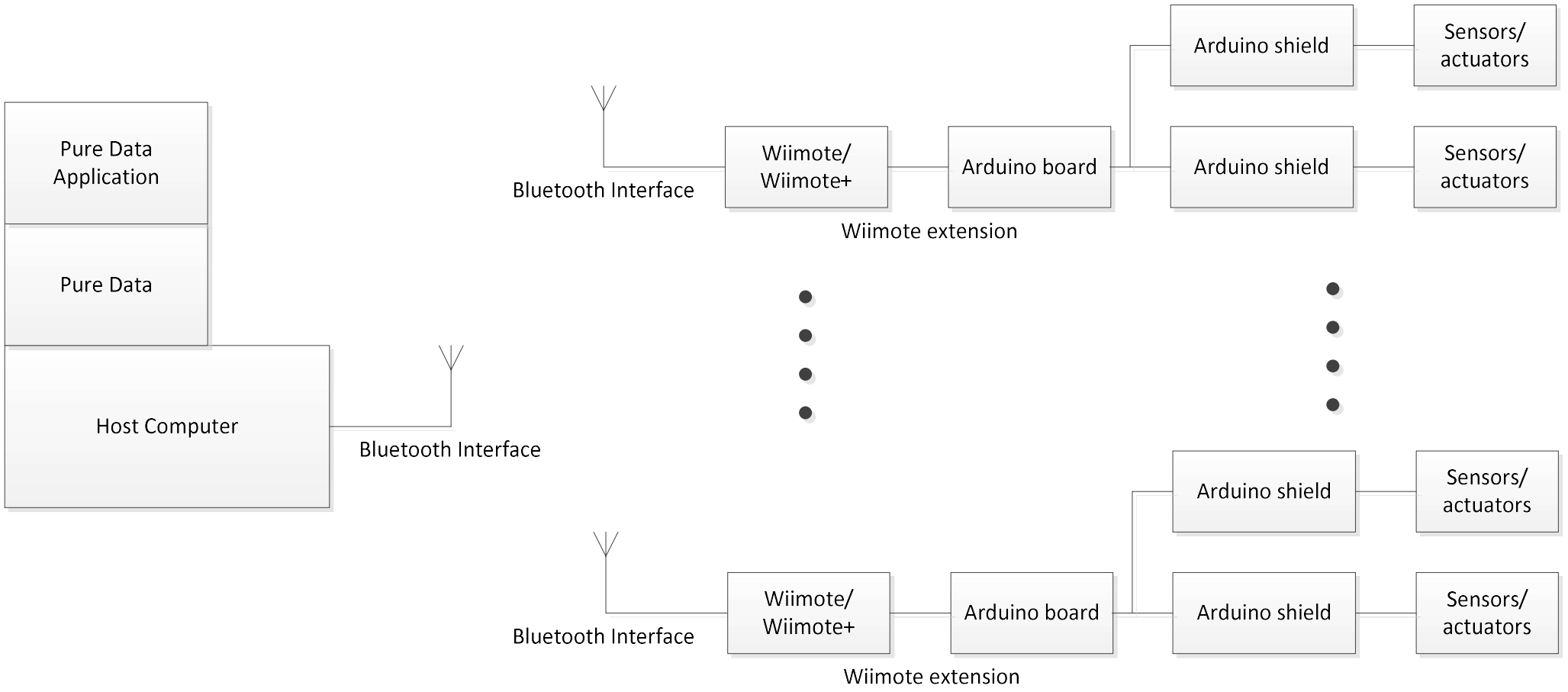

Pure Data Application
OSC
Pure Data
USB
PC (Windows, MacOS, Linux)
Inputs
Outputs
Control logic
OSC
Arduino

- Arduino Mega
- Wiimote/Wiimote+
- LED (NeoPixels/WS2812)
- Bluetooth speaker
- Bluetooth mic
- Pure Data on PC
  - Communicate with Arduino Mega over Bluetooth through Wiimote/Wiimote+





Whisper: Message in the Bottle. https://youtu.be/K56zk-whHWE


Pure Data Application
Pure Data
Host Computer
Bluetooth Interface
Bluetooth Interface
Wiimote/ Wiimote+
Arduino board
Arduino shield
Sensors/ actuators
Arduino shield
Sensors/ actuators
Wiimote extension
Bluetooth Interface
Wiimote/ Wiimote+
Arduino board
Arduino shield
Sensors/ actuators
Arduino shield
Sensors/ actuators
Wiimote extension

- Arduino UNO
- Capacitive touch sensor
- Servo motor, off-center weight for rocking
- LEDs (NeoPixel/WS2812)
- Bluetooth module (HC-05)
- MIT App Inventor on Android phone
  - Communicate with Arduino UNO via HC-05



https://youtu.be/0o1elEBKH5I

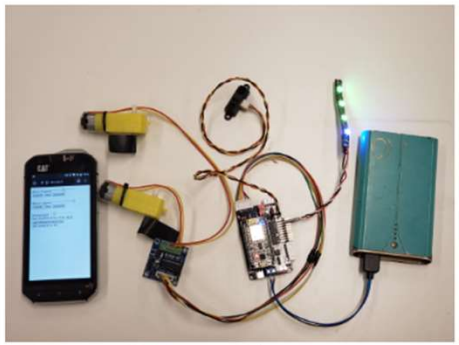
- Web Server running on ESP8266
- Two DC motors, driven by a two-channel motor driver (L293D)
- LEDs (NeoPixel/WS2812)
- IR distance sensor (Sharp GP2Y0A21)
- Circuit powered by a power bank (5V output)

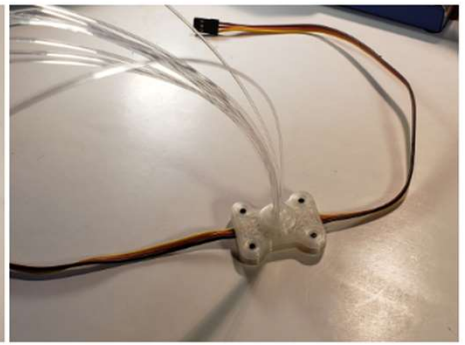
ESP8266 Base Controller Board and Some Sensors and Actuators

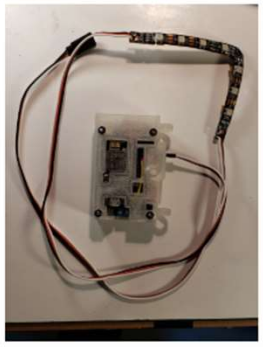
Created by Choy Sze Tsan Clifford, last modified on Jan 25, 2026 10:25

The ESP8266 Base Controller Board with

1. a strip of NeoPixels (WS2812)
2. two motors, driven by L293d driver board





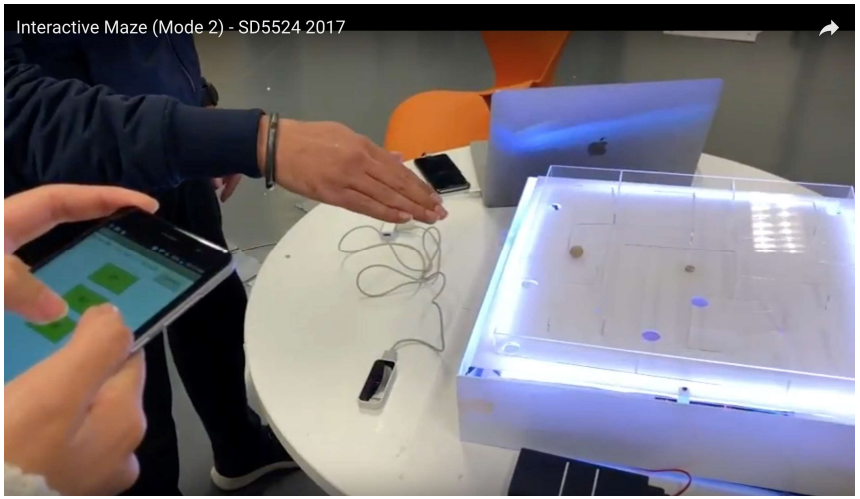


https://www.designandmake.org/x/iRCwEQ

- Leap Motion connected to a PC
- Software running on PC
  - Get hand gesture from Leap Motion
  - Send message to ESP8266 to control two servo motors (via websocket interface)
- ESP8266
- Servo motors with pan-tilt mechanism
- Servo motor for turning multiple “gates” to affect ball movement
- MIT App Inventor to control the “gates”

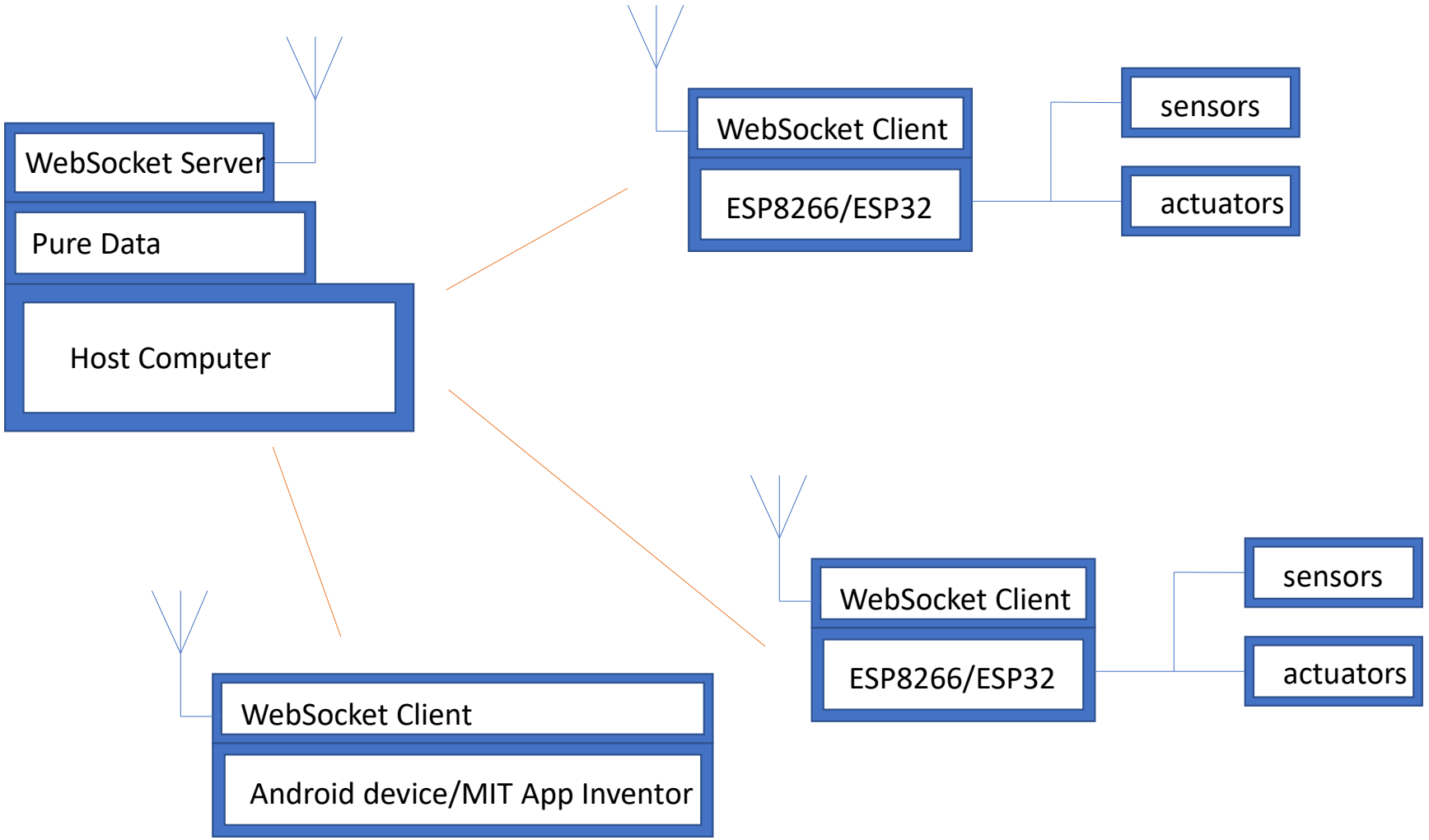


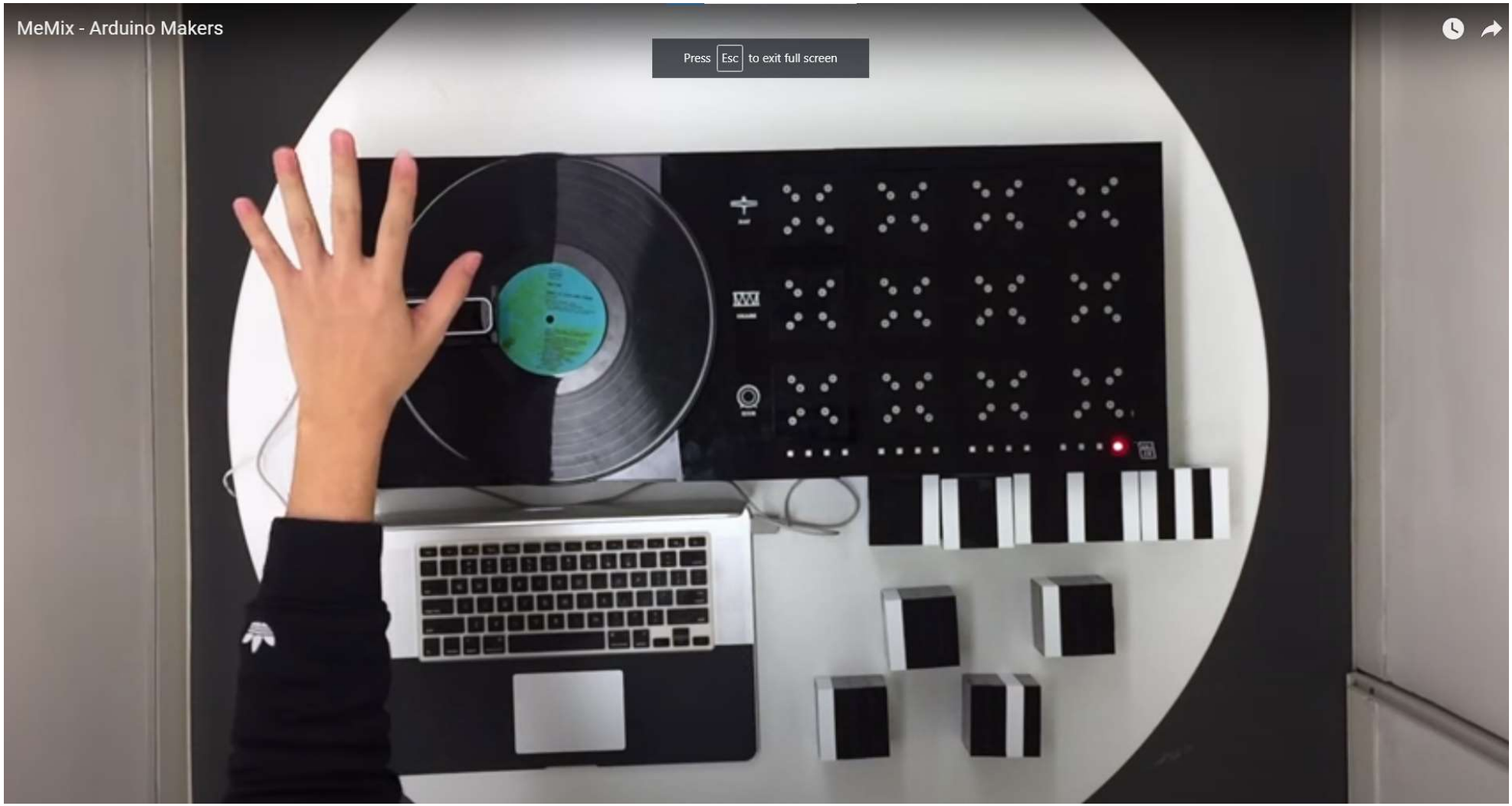
https://youtu.be/OGYn3Ugqo0I



https://youtu.be/uvPXa6h9aX0

https://www.designandmake.org/display/shareit/76222397/DKKceff236f49c44502a83ef5f64e19c3c6VNH/%E6%88%91%E7%9A%84%E5%BD%B1%E7%89%87%2023.mp4?version=1


WebSocket Server
Pure Data
Host Computer
WebSocket Client
ESP8266/ESP32
sensors
actuators
WebSocket Client
ESP8266/ESP32
sensors
actuators
WebSocket Client
Android device/MIT App Inventor



https://youtu.be/r0s-M0i3vLk



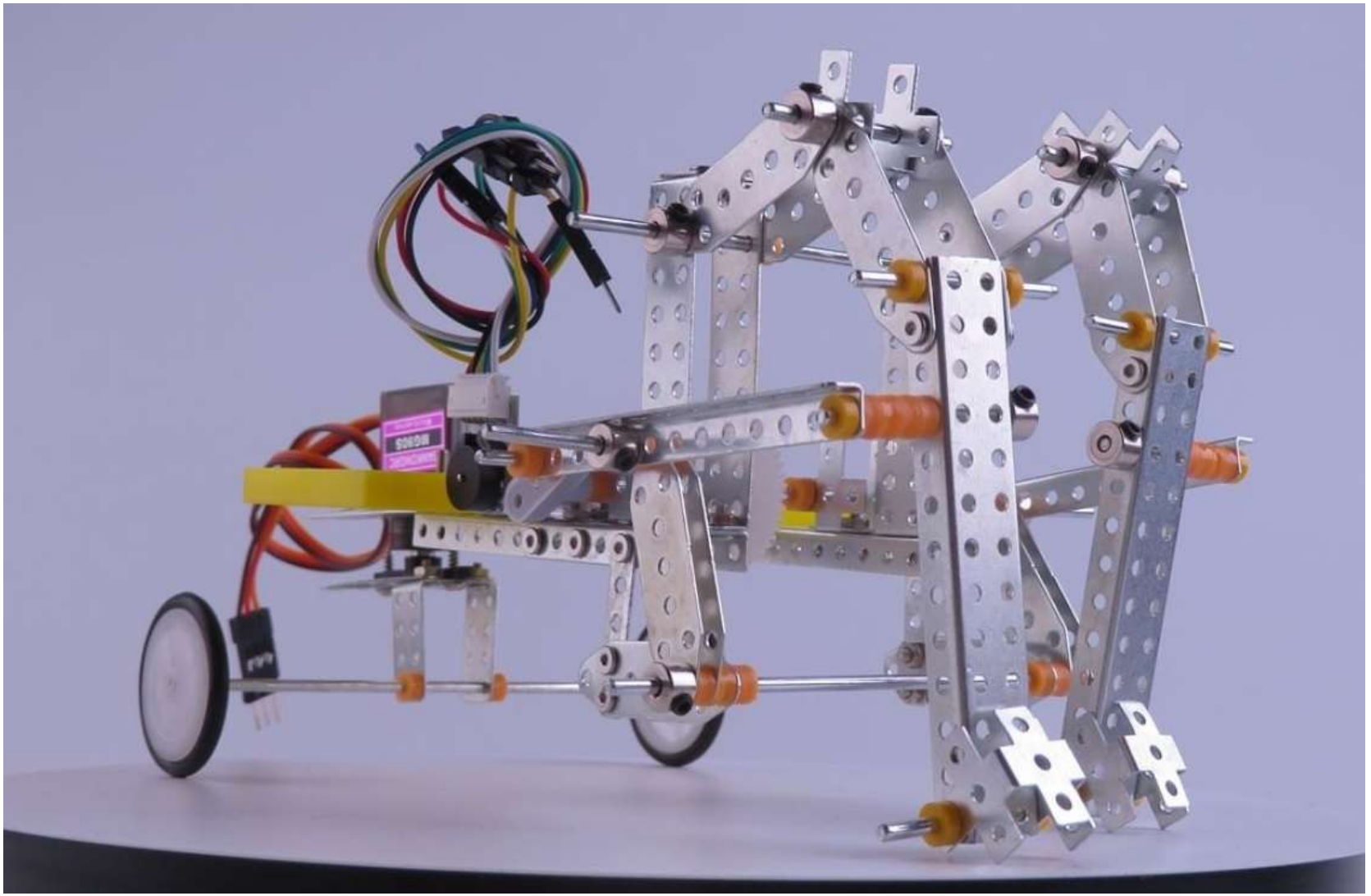
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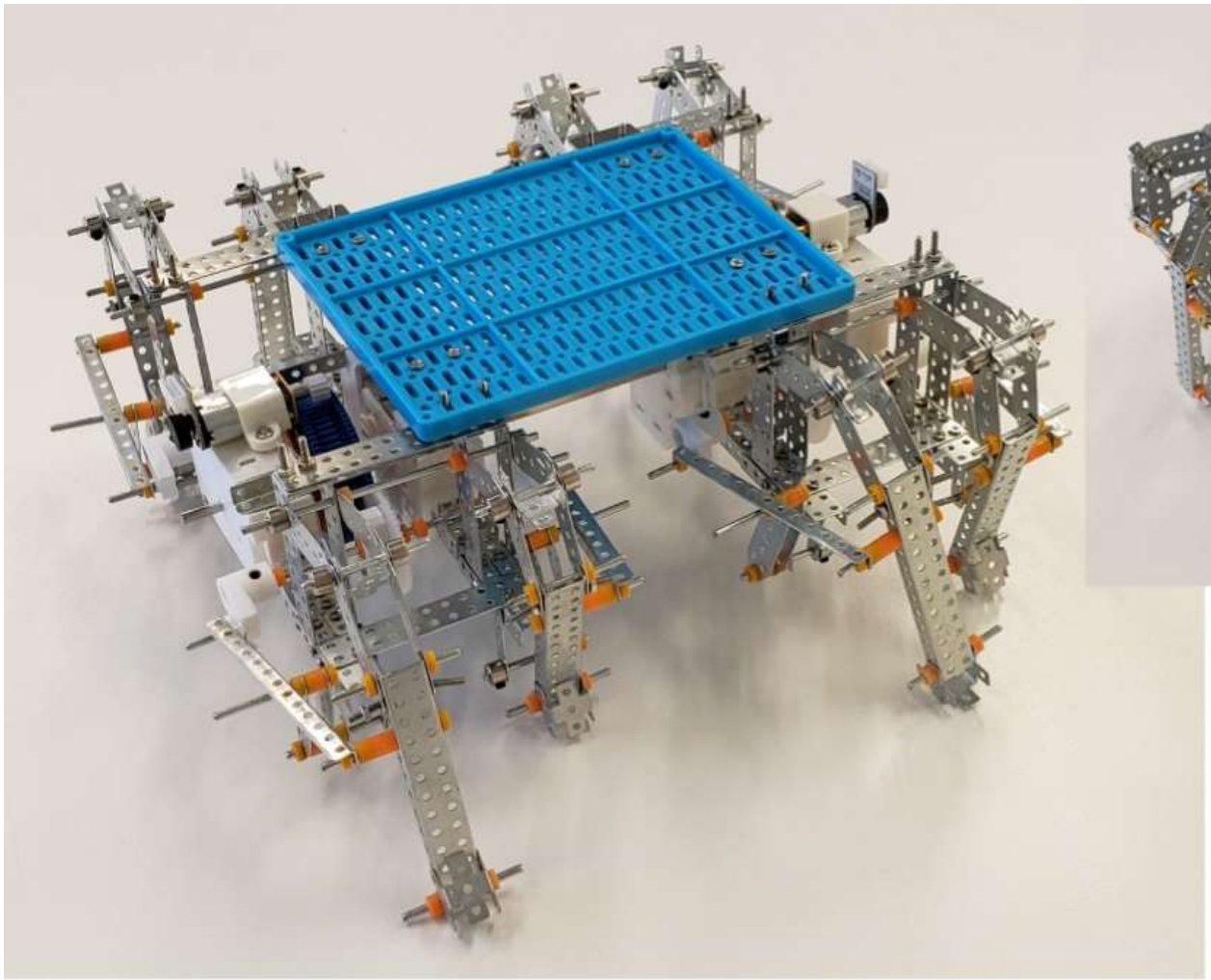
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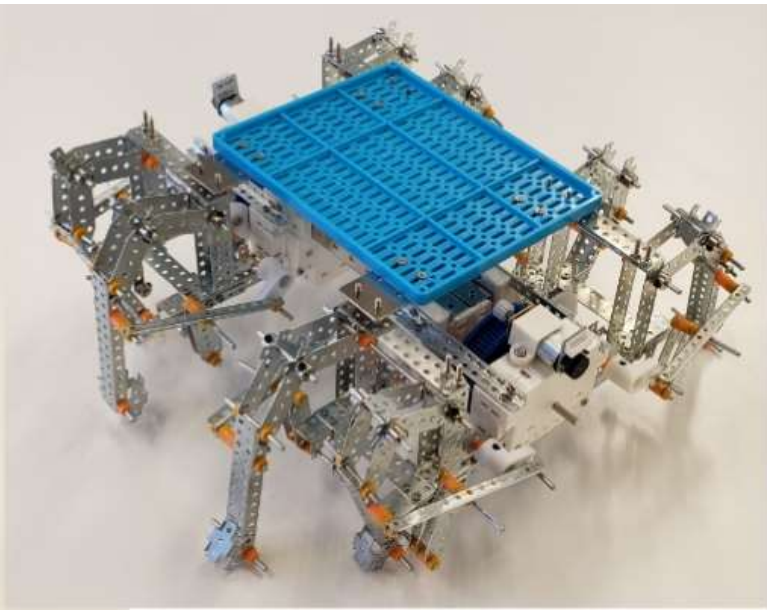

CENTRAL CLASSROOM
8-Legged Walker Robot Workshop
CENTRAL CLASSROOM
Building a Railway-connected Community Workshop
31/10/2021


PolyU Design
MATCH POINT Co.
Coordinator
Supported by
STEM YOUTH
in alphabetical order only
CENTRAL CLASSROOM
8-Legged Walker Robot Workshop
10, 16, 17 & 30 /10/2021
CENTRAL CLASSROOM
Building a Railway-connected Community Workshop
31/10/2021
This event consists of three workshops with the aim to engage participants to create different elements for constructing a scaled model of a "Railway-Connected Community" together through hands-on activities. Through these workshops, participant will construct trains which can be controlled through mobile phone, buildings for the scaled model of the community as well as props representing people, pets, public furniture and other elements for the community.
CENTRAL CLASSROOM
Air Rockets Workshop
13 & 14 /11/2021
This workshop aims at engaging participants in construction of a paper rocket which is powered by compressed air, thereby arousing their interests in concepts in physics including forces and projectile motion. Participant will be creating a paper rocket with a piece of paper, and use a launcher to launch their air rocket



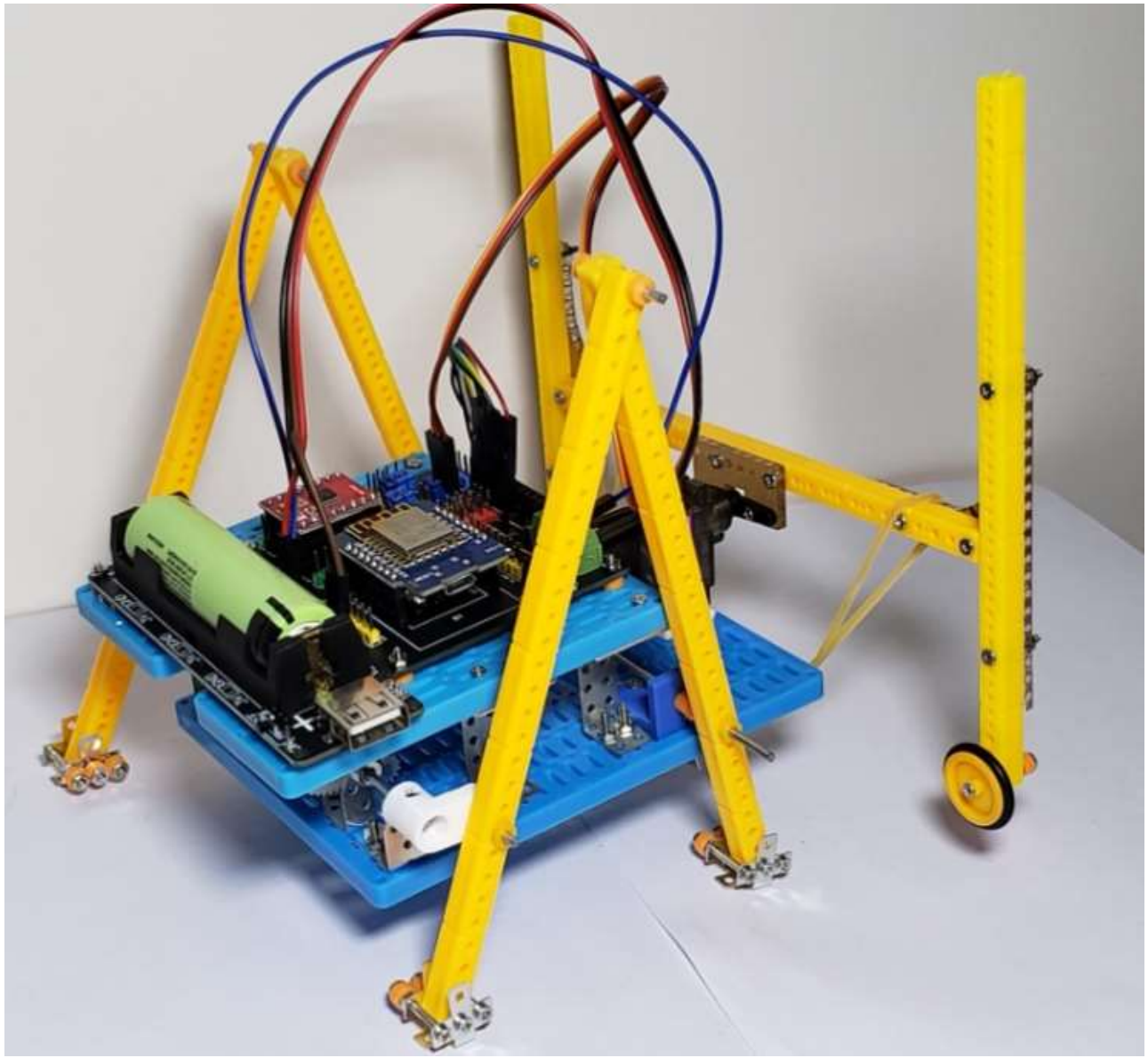
https://www.designandmake.org/x/jU6jBw

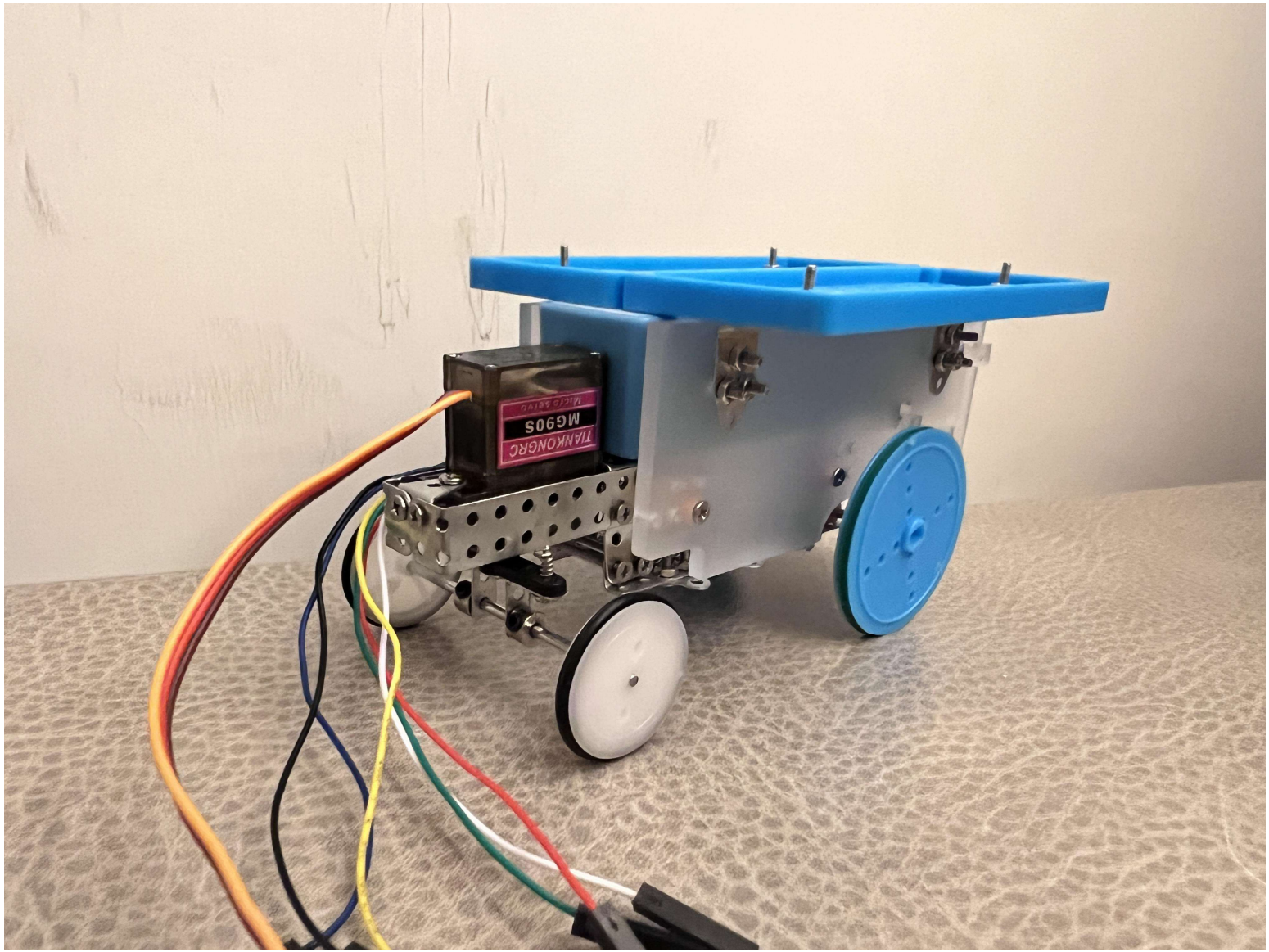




PolyU Design

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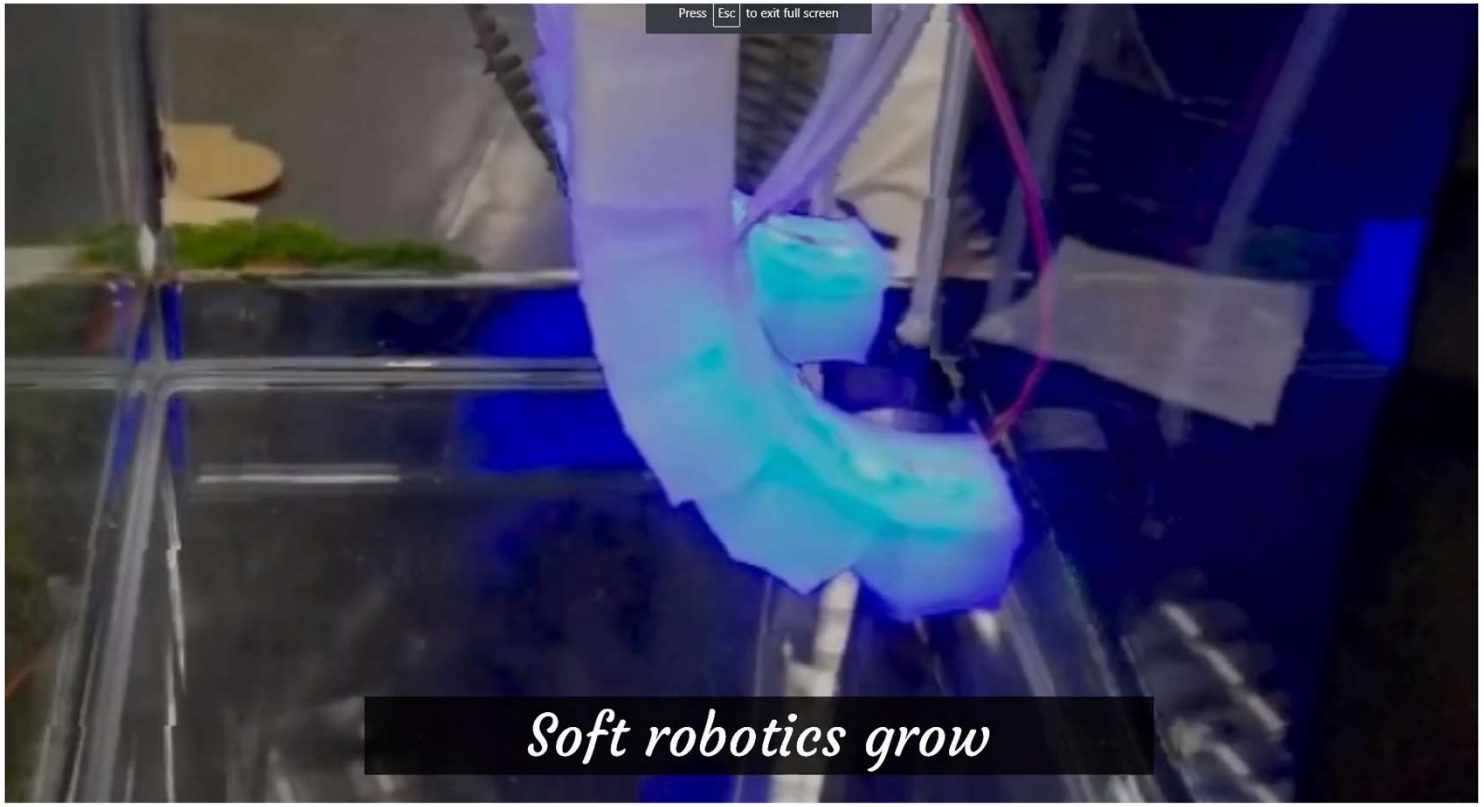

TIANKONGRC
MG90S

Mechanical Flowers



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Mechanical Flowers



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Mechanical Flowers



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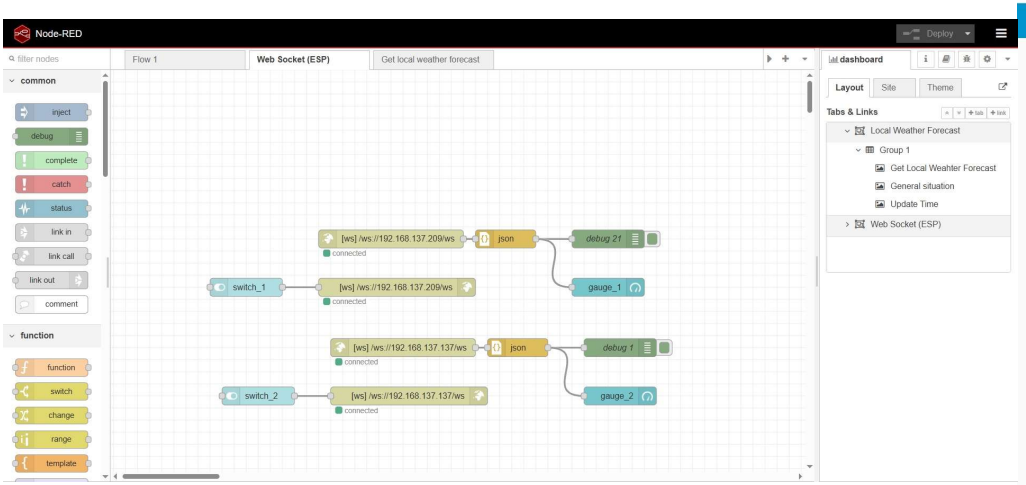
Some Technological Elements

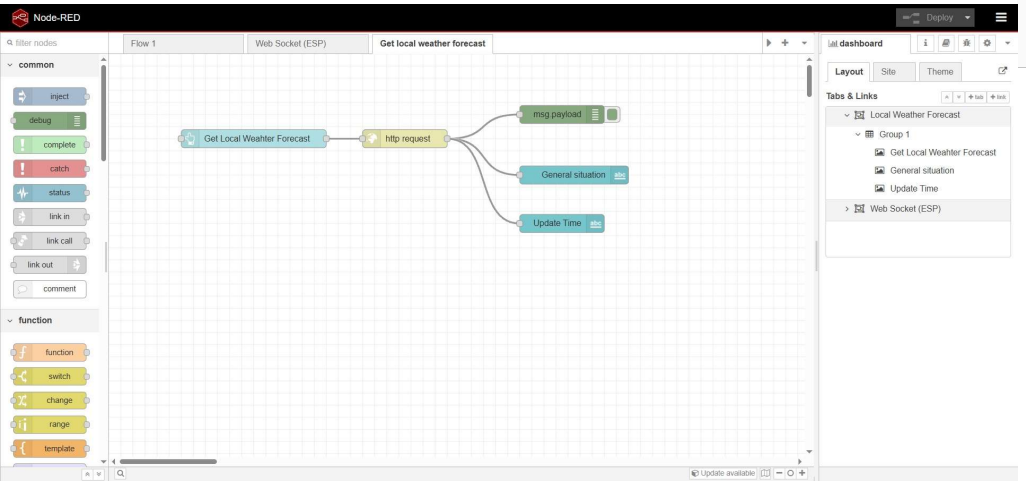
- Programming (C++, JavaScript)
- Arduino IDE
- ESP32 (ESP32C3)
- Electric circuit, electronics, interfacing
- Electro-mechanical devices, drivers
- Power converter
- Sensors, actuators
- Gear, linkage, …
- 3D printing

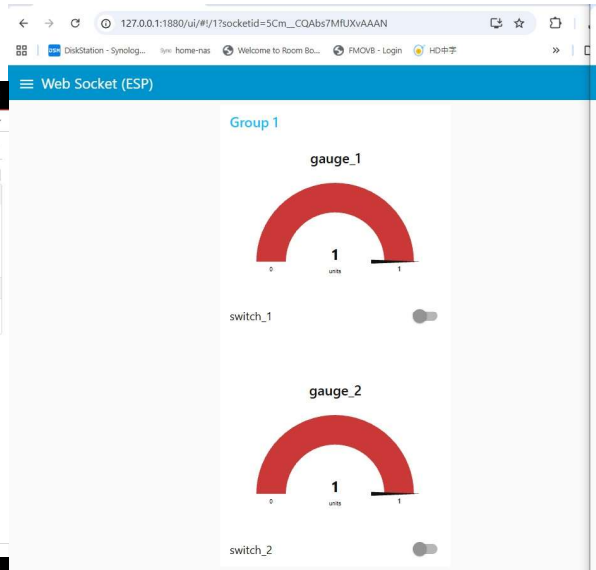
- Raspberry Pi
- NodeJS, Node-RED
- HTTP, WebSocket, MQTT
- JSON, RESTful API
- MIT App Inventor
- Pure Data (??)
- Rhino 3D, Grasshopper

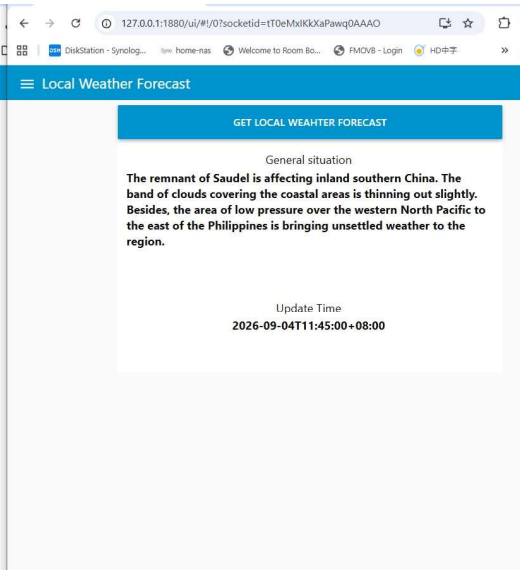
Focuses

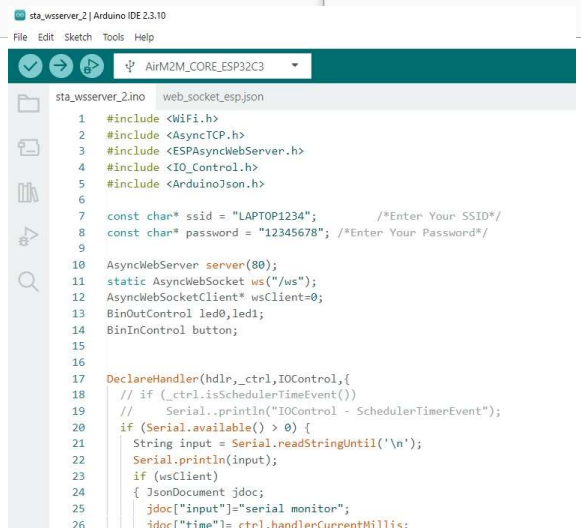
- Arduino (ESP32C3)
- Programming (basics to networking)
- Sensors and actuators
- Circuits and interfacing
- Power


Node-RED
Deploy
Flow 1
Web Socket (ESP)
Get local weather forecast
common
inject
debug
complete
catch
status
link in
link call
link out
comment
function
function
switch
change
range
template
[ws] /ws://192.168.137.209/ws
connected
json
debug 2
gauge_1
switch_1
[ws] /ws://192.168.137.137/ws
connected
debug 1
gauge_2
switch_2
dashboard
Layout
Site
Theme
Tabs & Links
Local Weather Forecast
Group 1
Get Local Weahter Forecast
General situation
Update Time
Web Socket (ESP)

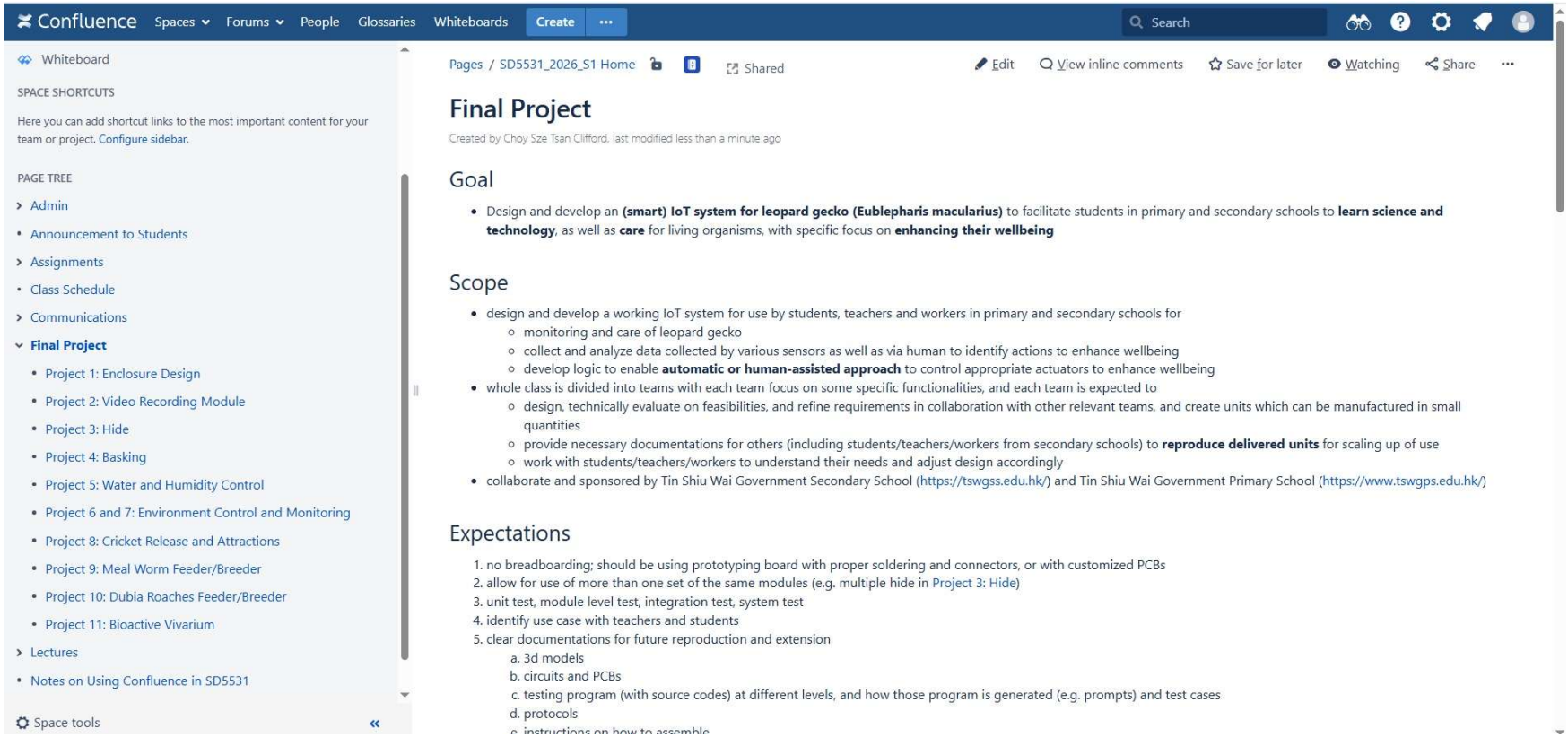

Node-RED
Deploy
Flow 1
Web Socket (ESP)
Get local weather forecast
common
inject
debug
complete
catch
status
link in
link call
link out
comment
function
function
switch
change
range
template
Get Local Weahter Forecast
http request
msg.payload
General situation
Update Time
dashboard
Layout
Site
Theme
Tabs & Links
Local Weather Forecast
Group 1
Get Local Weahter Forecast
General situation
Update Time
Web Socket (ESP)
Update available


127.0.0.1:1880/ui/#!/1?socketid=5Cm_CQAbs7MfUXvAAAN
Web Socket (ESP)
Group 1
gauge_1
1
units
switch_1
gauge_2
1
units
switch_2


127.0.0.1:1880/ui/#!/0?socketid=tT0eMxIKkXaPawq0AAAO
Local Weather Forecast
GET LOCAL WEAHTER FORECAST
General situation
The remnant of Saudel is affecting inland southern China. The band of clouds covering the coastal areas is thinning out slightly. Besides, the area of low pressure over the western North Pacific to the east of the Philippines is bringing unsettled weather to the region.
Update Time
2026-09-04T11:45:00+08:00


sta_wsserver_2 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
AirM2M_CORE_ESP32C3
sta_wsserver_2.ino
web_socket_esp.json
#include <WiFi.h>
#include <AsyncTCP.h>
#include <ESPAsyncWebServer.h>
#include <IO_Control.h>
#include <ArduinoJson.h>
const char* ssid = "LAPTOP1234"; /*Enter Your SSID*/
const char* password = "12345678"; /*Enter Your Password*/
AsyncWebServer server(80);
static AsyncWebSocket ws("/ws");
AsyncWebSocketClient* wsClient=0;
BinOutControl led0,led1;
BinInControl button;
DeclareHandler(hdlr,_ctrl,IOControl,{
// if (_ctrl.isSchedulerTimeEvent())
// Serial..println("IOControl - SchedulerTimerEvent");
if (Serial.available() > 0) {
String input = Serial.readStringUntil('\n');
Serial.println(input);
if (wsClient)
{ JsonDocument jdoc;
jdoc["input"]="serial monitor";

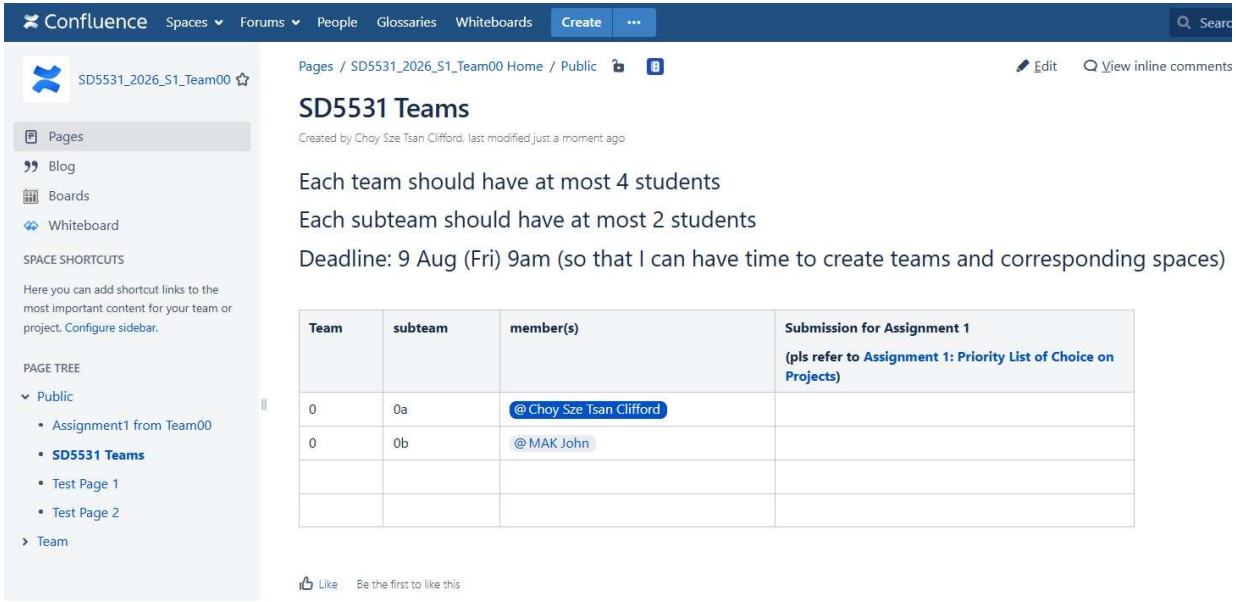
Final Project



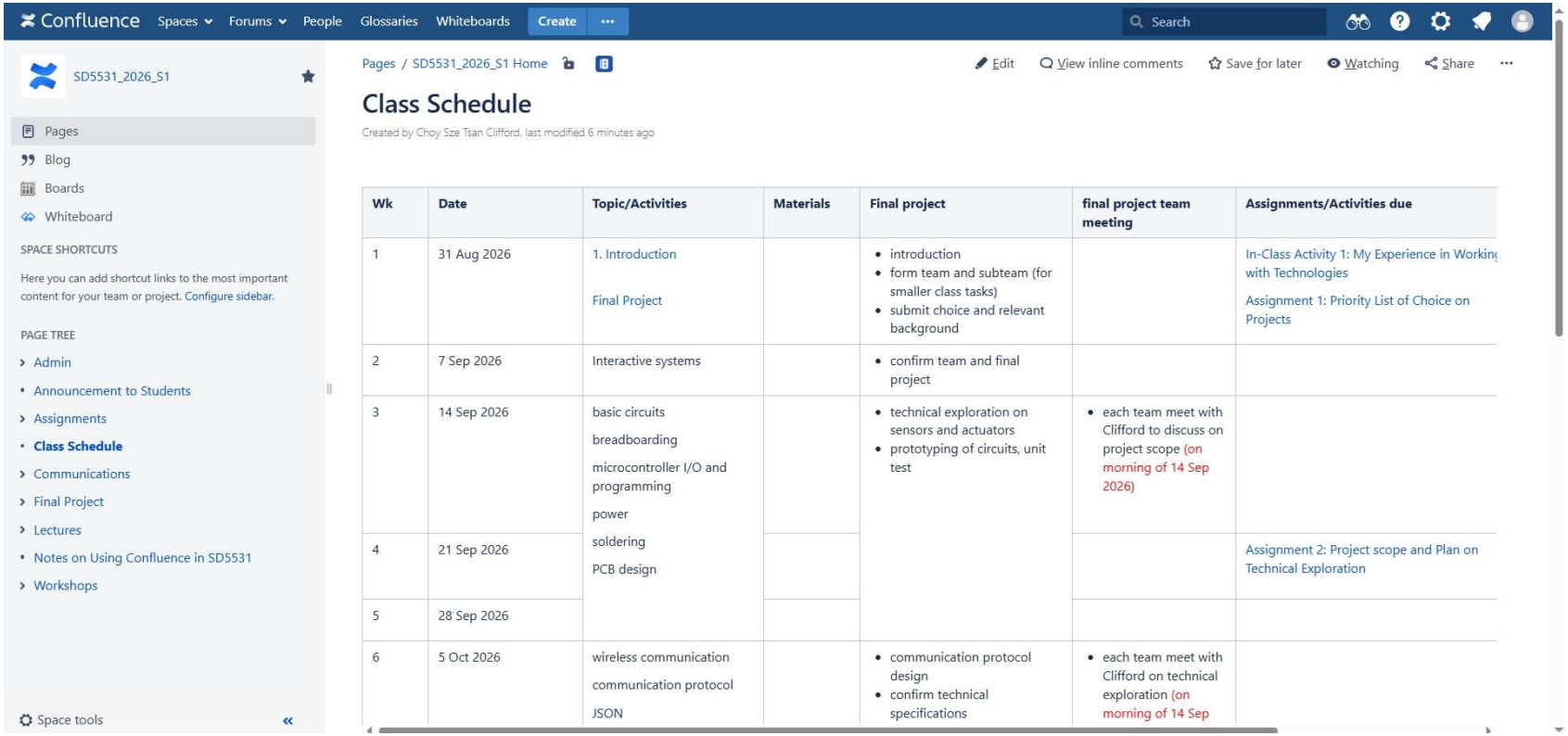
https://www.designandmake.org/x/NYT0Eg

Form Your Team

- 4 students at most per team
- 2 students at most per sub-team (for completing some workshop activities)
- Highly prefer your team members have different backgrounds (disciplinary knowledge, working experience, gender, …)
- Update https://www.designandmake.org/x/TIX0Eg



Schedule



Wk	Date	Topic/Activities	Materials	Final project	final project team meeting	Assignments/Activities due
1	31 Aug 2026	1. Introduction Final Project		• introduction • form team and subteam (for smaller class tasks) • submit choice and relevant background		In-Class Activity 1: My Experience in Working with Technologies Assignment 1: Priority List of Choice on Projects
2	7 Sep 2026	Interactive systems		• confirm team and final project		
3	14 Sep 2026	basic circuits breadboarding microcontroller I/O and programming power soldering PCB design		• technical exploration on sensors and actuators • prototyping of circuits, unit test	• each team meet with Clifford to discuss on project scope (on morning of 14 Sep 2026)	
4	21 Sep 2026					Assignment 2: Project scope and Plan on Technical Exploration
5	28 Sep 2026					
6	5 Oct 2026	wireless communication communication protocol JSON		• communication protocol design • confirm technical specifications	• each team meet with Clifford on technical exploration (on morning of 14 Sep	

https://www.designandmake.org/x/uoT0Eg

Note on Schedule

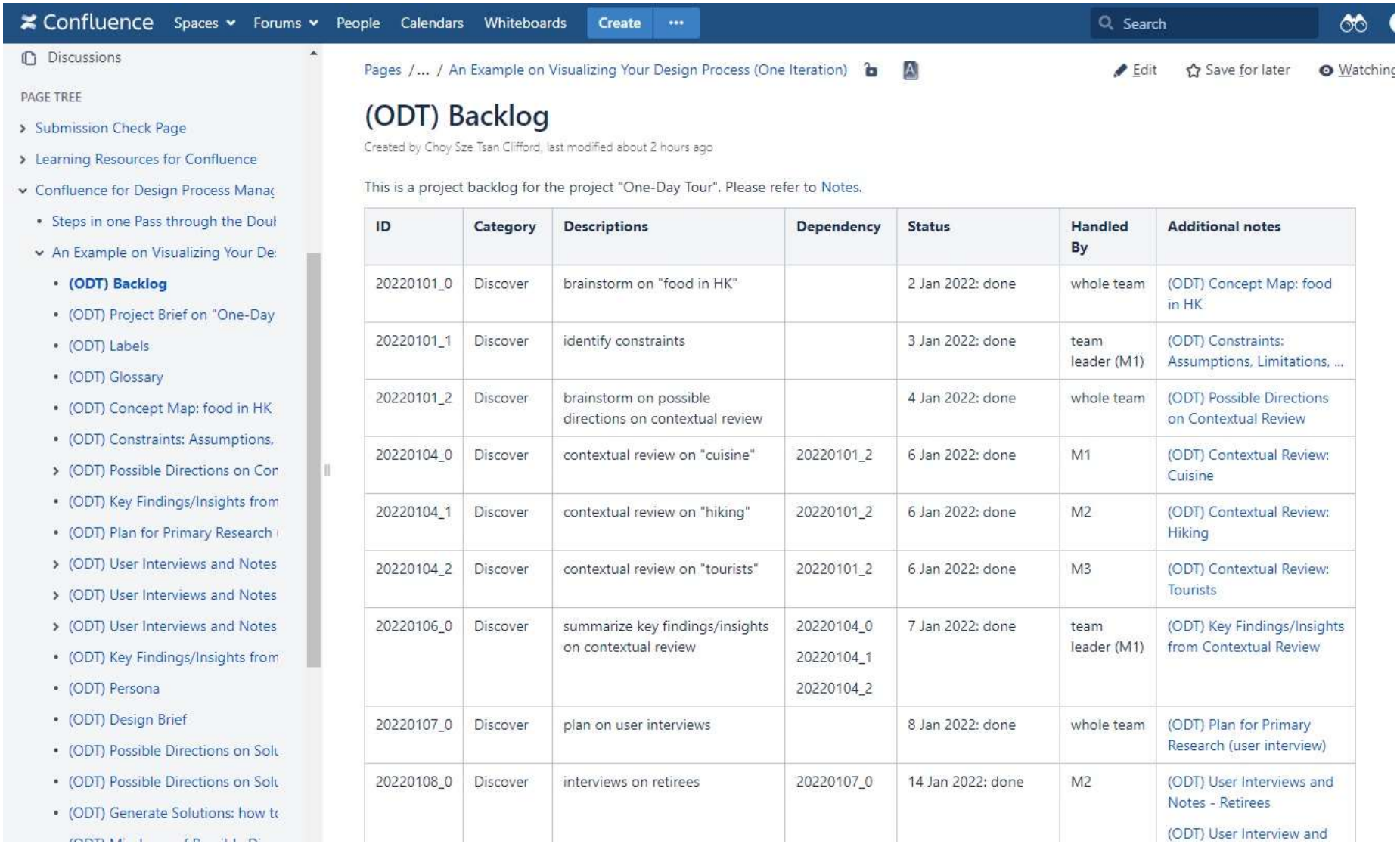
- Availability on some Fri morning
  - Soldering workshop
  - Tutorials
- Availability on some Sat morning
  - Work with secondary school students and teachers
- May have to visit TSWGSS/TSWPSS
  - E.g. understand space requirements, understand users' needs, understand other constraints, test with real gecko

Assessment

- In-class activities (30%)
- Final Project (70%)
  - Assignments 2 to 6

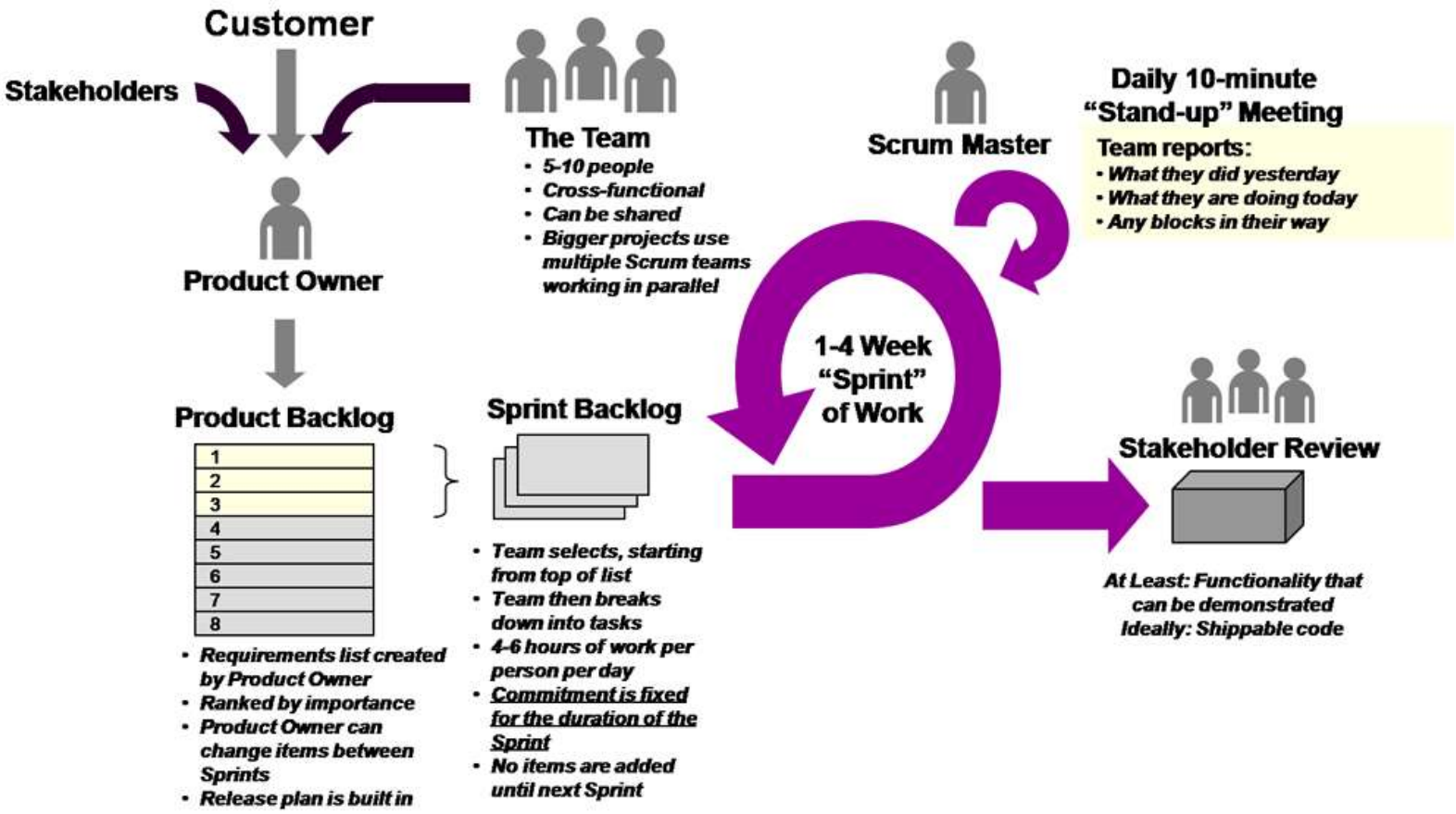
Teamwork and Development Log

- Teamwork
  - Independent and Interdependent
- Development Log
  - Update regularly, including success and failures
  - Evidence to demonstrate "independency" and "interdependency"
  - Evidence on individual contributions, and identify free-riders
  - Provide support on your individual contributions in "Individual Reflection Report"
  - Keep in your team space, under the page "Team"
- Examples:
  - https://www.designandmake.org/display/shareit/221611485/ZVOaa62bed75c414a6e951071728bdcd1b4WNN
  - https://www.designandmake.org/x/EIX8EQ
  - https://www.designandmake.org/x/GQKKEQ



https://www.designandmake.org/display/TUTORIAL/%28ODT%29+Backlog

Agile Scrum Process



http://pmtips.net/adapting-agile-methodology-startup/

Use of GenAI

- Written submissions
  - Refer to https://www.polyu.edu.hk/ar/students-in-taught-programmes/use-of-genai/
- Project related background knowledge, technical knowledge, coding
  - Use genAI to facilitate learning and understanding, not just generate something without understanding how it works
  - Check for accuracy
- Image, video
  - Facilitate communications

Things to do

- Form your team, and update table in https://www.designandmake.org/x/TIX0Eg
- Go through projects descriptions in https://www.designandmake.org/x/NYT0Eg, search for all those terms mentioned, and put any questions in comments
- Work on Assignment 1 and update table in https://www.designandmake.org/x/TIX0Eg
- Check out those sensors and actuators in
  - https://www.designandmake.org/display/DET4M/Sensors+or+Detectors
  - https://www.designandmake.org/display/DET4M/Effectors+or+Actuators